Amr Abdalla

Montréal, QC | 438-527-2478 | amrala304@gmail.com | linkedin.com/in/amr-a-abdalla | amr-abdalla.github.io/portfolio-website

Summary

Gameplay programmer with 4+ years of Unity/C# experience building ability and stat systems across three games, credited on *Last Epoch* (Eleventh Hour Games) and *Last Loremaster*. Completing a Master's in Software Engineering at Concordia. Eligible for an open work permit (PGWP) on graduation, with no employer sponsorship required; open to relocation in Canada.

Professional Experience

Gameplay Programmer Oct. 2025 – May 2026
WhySo Studios · Dark Star, rogue-lite fleet-command strategy · Contract *Montréal, QC*
- Extended the ship-systems interface (munitions, hangars) with a third category, abilities, from stat buffs to area-effect fields
- Refactored ship stats into composable effect modifiers each ship adds and removes independently; built the first tutorial system

Teaching Assistant, Graduate Software Engineering Sept. 2025 – Present
Concordia University · Part-time *Montréal, QC*
- Labs, grading, office hours for four terms: Software Comprehension & Maintenance, Software Measurement, Intelligent Systems

Gameplay Programmer June 2021 – May 2025
The Bad Nomads, co-development partner to Eleventh Hour Games · Last Epoch, Last Loremaster *Remote*
- Implemented, with its designer, the "Targeting Paradigm" that overhauled *Last Epoch*'s gamepad skill targeting: per-skill behaviors (cast on self, closest enemy, forward cone) that adapt to skill-tree nodes and gear; shipped in Season 2 (patch notes)
- One year inside EHG's 20+ engineer team: reported to EHG leads, code-reviewed across studios, onboarded two developers
- One of three programmers on *Last Loremaster*, alongside a designer and artists, from inception to Steam demo: owned crafting, skill modifiers, and unique items; co-designed the UI architecture; built player and enemy skills
- Took the *Last Loremaster* dev build from 15 to 60 FPS: traced the slowdown with the Unity Profiler to stats recomputing every frame, then rewrote the invalidation logic so they recompute only on change

Game Designer and Programmer June 2022 – June 2024
The American University in Cairo, with Megawra (heritage NGO) · Part-time *Cairo, Egypt*
- Designed and built two educational Unity arcade mini-games, deployed on a custom cabinet, playtested with 30–60 children
- Wrote a shader that makes a 2D mesh behave as water waves, plus a script that procedurally generates those meshes

Research and Projects

PEAK — Automated Playtesting Engine | *Code & Sorcery Lab, Ontario Tech University* | GitHub Jan. 2026 – Present
- Built the stats layer and dashboard of PEAK, a Python tool that evolves neural networks to playtest 2D levels for designers
- Designed its stats observer so new metrics are added by editing a YAML file, with no changes to game code: recorders attach to engine methods via reflection (aspect-oriented approach, written up on the Ptidej Team blog)

Education

Concordia University Montréal, QC
Master of Engineering in Software Engineering, GPA 4.15/4.30 *May 2025 – Dec. 2026*
- Advanced Game Development (COMP 6331): A+

The American University in Cairo Cairo, Egypt
Bachelor of Science in Computer Engineering, GPA 3.38/4.00 *2018 – 2024*
- Completed part-time over six years while working full-time as a gameplay programmer

Technical Skills

Languages: C#, C++, Python, Java, HLSL/ShaderLab
Game Development: Unity, gameplay systems, controller abstraction, ability and stat systems, UI architecture, shaders, profiling
Tools & Practice: Git, code review, Jira, ClickUp, Unity Profiler, design patterns
Spoken Languages: English (fluent), Arabic (native), French (beginner)

Extra-Curricular

First Place, Reanimate Summer School Game Jam | *Escape from Reanimate* | crvja.reanimate.school June 2026
- Gameplay programmer on a team of four; built an escape-room game where players find items and clues to escape

Founder and Mentor, Game Development Track | *Google Developer Student Club, AUC* Jan. – Mar. 2023
- Created and taught a five-session introductory course on Unity, design patterns, and shaders